EVERY NONEMPTY LOCALLY CLOSED SUBSET OF AN ALGEBRAIC SPACE MEETS THE FINITE TYPE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement includes a displayed implication and an 'in particular' sentence; both are inside the counted statement span.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The reduced induced subspace structure on a locally closed subset. Remark remark-space-structure-locally-closed-subset, used at proof lines 5144-5146. Produces the algebraic space Z with $|Z| = T$.
- Immersions are locally of finite type. Lemma lemma-immersion-locally-finite-type, used at proof lines 5147-5148. Makes Z -> X locally of finite type.
- Finite type points behave well under locally finite type morphisms. Lemma lemma-finite-type-points-morphism, used at proof lines 5149-5151. Gives that the finite type points of Z lie among those of X.
- Nonempty affine schemes have closed points and finite type points of a space come from them. Lemma lemma-identify-finite-type-points, used at proof lines 5152-5155. Produces a finite type point of Z.

Source reference. Original lemma 06EI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8217–8217. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06EH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8216–8216. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8212–8212. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 06EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8211–8211. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. For any locally closed subset $T \subset |X|$ we have*

$$T \neq \emptyset \Rightarrow T \cap X_{ft\text{-}pts} \neq \emptyset.$$

In particular, for any closed subset $T \subset |X|$ we see that $T \cap X_{ft\text{-}pts}$ is dense in T.

Proof. Let $i : Z \to X$ be the reduced induce subspace structure on T, see Remark 06EC. Any immersion is locally of finite type, see Lemma 06ED. Hence by Lemma 06EI we see $Z_{\text{ft-pts}} \subset X_{\text{ft-pts}} \cap T$. Finally, any nonempty affine scheme U with an étale morphism towards Z has at least one closed point. Hence Z has at least one finite type point by Lemma 06EH. The lemma follows. □